

Draft - Requires Review

Protocol PROT-2026-014 - HX-100 Heat Exchanger Mounting Bracket

Project information and fixed statements

[Process control: deterministic / template-based]

Protocol ID: PROT-2026-014

Title: Torque retention verification

Component: HX-100 Heat Exchanger Mounting Bracket

Protocol revision: B

Protocol date: 2026-10-01

Test method: TQ-7 Torque Retention Test

Acceptance criterion: TQ-7 Torque Retention Test result must be within 45 to 55 Nm (all samples within range).

Attachments supplied with this report:

- Attachment A: equipment_calibration_record.pdf

Test method and reference guidance

[Process control: RAG - retrieval over reference documents, sources cited]

The following statements are retrieved verbatim from the supplied reference documents; each is cited to its source file and location.

- Reports in this programme follow a fixed structure: project information, test method and reference guidance, results and evidence, then summary and observations. Every results section must state the acceptance range and show a per-sample Pass/Fail verdict. [Reporting_guidance.md para1]
- A calibrated torque wrench with a current calibration record is required. The wrench must cover the expected range with margin, and its calibration sticker must be checked before the first measurement of the day. [TQ-7_method_summary.md para2]
- The acceptance limits are defined by the protocol, not by this method document. A sample passes when its measured torque lies inside the protocol range, inclusive of the limits. Any result outside the range is a failure and must be reported as measured without adjustment. [TQ-7_method_summary.md para4]
- This method verifies that the tested fasteners retain the specified torque after the conditioning cycle. The result of each sample is a single torque value in newton-metres read at the moment of first movement. [TQ-7_method_summary.md para1]

Sources: Reporting_guidance.md para1; TQ-7_method_summary.md para2; TQ-7_method_summary.md para4; TQ-7_method_summary.md para1

Results and evidence

[Process control: evidence-based, deterministic processing]

Raw data source: results.csv (5 rows), evaluated against the protocol acceptance range [45, 55] Nm.

Pass/Fail is calculated deterministically by the application; no AI involvement.

- S-01: 48.2 Nm -- Pass (48.2 within [45, 55]) [results.csv row 1]
- S-02: 51.7 Nm -- Pass (51.7 within [45, 55]) [results.csv row 2]

- S-03: 46.9 Nm -- Pass (46.9 within [45, 55]) [results.csv row 3]
- S-04: 53.1 Nm -- Pass (53.1 within [45, 55]) [results.csv row 4]
- S-05: 49.8 Nm -- Pass (49.8 within [45, 55]) [results.csv row 5]

Summary: 5 samples, 5 passed, 0 failed.

Sample ID	Measurement	Unit	Verdict	CSV row
S-01	48.2 Nm	Nm	Pass	row 1
S-02	51.7 Nm	Nm	Pass	row 2
S-03	46.9 Nm	Nm	Pass	row 3
S-04	53.1 Nm	Nm	Pass	row 4
S-05	49.8 Nm	Nm	Pass	row 5

Summary and observations

[Process control: deterministic / template-based]

A total of 5 samples were evaluated against the protocol acceptance range [45, 55] Nm: 5 passed and 0 failed. All measured results were within the acceptance range. This summary is a draft generated from verified results and retrieved guidance; review and edit before approval.

Note: Generated without AI; fully deterministic.

Sources: Reporting_guidance.md para1; Measurement_practice_notes.md para1; TQ-7_method_summary.md para4

Generated Excel table (results_table.xlsx)

Readable rendering of the generated workbook - identical values to the attached XLSX file.

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Protocol PROT-2026-014: HX-100 Heat Exchanger Mounting Bracket -- evaluated against [45.0, 55.0] Nm		
Sample ID	Measurement	Unit
S-01	48.2	Nm
S-02	51.7	Nm
S-03	46.9	Nm
S-04	53.1	Nm
S-05	49.8	Nm
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Metric	Value	
Protocol ID	PROT-2026-014	
Component	HX-100 Heat Exchanger Mounting Bracket	
Total samples	5	
Passed	5	
Failed	0	
No data	0	
Acceptance range	45 to 55 Nm	

Attachment A

equipment_calibration_record.pdf

Equipment Calibration Record (fictional)

Equipment: TQ-W-12 torque wrench, serial TW-88231, range 10-60 Nm

Calibrated by: Metrology Ltd (fictional vendor)

Calibration date: 2026-09-10

Next due: 2027-03-10

Result: PASS - as-found deviations within manufacturer specification.

Statement: This record is a synthetic sample document created for a report-generator prototype. It is not a real calibration certificate.